

Companion XXXI: The RDCC Halo Model and the Architecture of the Non-Linear Universe

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Abstract

This Companion develops a comprehensive non-linear structure formation framework for the Relaxation-Driven Cyclic Cosmology (RDCC). While earlier Companions concentrated on linear and quasi-linear observables, the present work examines the transition to fully non-linear scales, the formation and evolution of dark matter halos, and the resulting architecture of the cosmic web. We derive the RDCC-modified spherical-collapse threshold, the halo mass function, bias relations, and the non-linear matter power spectrum. The relaxation-induced suppression of small-scale potential evolution produces characteristic signatures in halo abundance, clustering strength, and internal halo structure. This Companion thereby establishes the theoretical foundation for non-linear cosmology within RDCC and lays the groundwork for future studies of galaxy formation, cluster physics, and cosmic web topology.

1 Introduction

The non-linear Universe is shaped by the formation and evolution of dark matter halos — the fundamental building blocks of galaxies, clusters, and the cosmic web. In standard cosmology, halo formation is governed by the linear growth factor, the matter power spectrum, and the spherical-collapse threshold. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified evolution of gravitational potentials introduces a scale-dependent suppression of small-scale growth together with an enhancement of large-scale coherence.

This Companion develops a unified RDCC halo model that connects linear perturbation theory with the fully non-linear regime. The results extend the linear framework of Companions XVI–XIX and provide a consistent description of non-linear matter clustering, halo statistics, and the large-scale architecture of the cosmic web.

2 From Linear to Non-Linear Structure in RDCC

The linear matter power spectrum in RDCC is modified by the relaxation scale k_{rel} :

$$P_{\text{lin,RDCC}}(k, a) = P_{\text{lin}}(k, a) \left[1 - \chi_P \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (1)$$

Because the growth rate itself becomes scale-dependent, the mapping from linear to non-linear scales is altered. We define the dimensionless power as

$$\Delta^2(k) = \frac{k^3}{2\pi^2} P(k). \quad (2)$$

In RDCC the non-linear power spectrum is obtained via a relaxation-modified halo-model mapping:

$$\Delta_{\text{NL}}^2(k) = \mathcal{F}_{\text{RDCC}}(\Delta_{\text{lin}}^2(k)). \quad (3)$$

The most important consequences are a suppression of non-linear power on small scales, an enhancement of coherence on large scales, a clear turnover at k_{rel} , and overall smoother transitions between the linear and non-linear regimes.

3 Halo Collapse in the RDCC Framework

The spherical-collapse threshold itself becomes scale-dependent because of the modified potential evolution:

$$\delta_{c,\text{RDCC}}(k) = \delta_c \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

This modification has several important consequences: massive halos tend to form earlier, the collapse of low-mass halos is delayed, small-scale fragmentation is reduced, and halo density profiles become smoother overall. The virial overdensity is affected in a similar way:

$$\Delta_{\text{vir,RDCC}} = \Delta_{\text{vir}} \left[1 - \chi_{\text{vir}} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

4 The RDCC Halo Mass Function

The Press–Schechter mass function is modified accordingly:

$$n_{\text{RDCC}}(M) = n(M) \left[1 - \chi_n \left(\frac{k(M)}{k_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

As a result, RDCC predicts an enhanced abundance of massive halos, a suppression of low-mass halos, a characteristic turnover mass M_{rel} , and generally smoother slopes of the mass function.

5 Halo Bias in RDCC

The halo bias takes the form

$$b_{\text{RDCC}}(M) = b(M) \left[1 - \chi_b \left(\frac{k(M)}{k_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

This scale-dependent bias leads to stronger large-scale clustering, weaker small-scale clustering, redshift-dependent bias signatures, and tracer-dependent clustering amplitudes.

6 Non-Linear Matter Power Spectrum in RDCC

Within the halo model the total non-linear matter power spectrum is the sum of the one-halo and two-halo contributions:

$$P_{\text{NL}}(k) = P_{1h}(k) + P_{2h}(k). \quad (8)$$

In RDCC both terms are modified:

$$P_{1h,\text{RDCC}}(k) = P_{1h}(k) \left[1 - \chi_{1h} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (9)$$

$$P_{2h,\text{RDCC}}(k) = P_{2h}(k) \left[1 - \chi_{2h} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (10)$$

Consequently, the non-linear power is suppressed on small scales, enhanced on large scales, exhibits a turnover at k_{rel} , and shows smoother transitions between linear and non-linear regimes.

7 Implications for the Cosmic Web

These modifications propagate directly into the large-scale morphology of the cosmic web. Filaments become thicker, voids grow larger, the connectivity between halos and filaments becomes smoother, and small-scale substructure is reduced. Such signatures offer promising observational tests for RDCC through weak lensing, galaxy clustering, and cosmic web statistics.

8 Conclusion

This Companion establishes a consistent halo model for the Relaxation-Driven Cyclic Cosmology and demonstrates its implications for non-linear structure formation. The relaxation-modified evolution of gravitational potentials fundamentally alters halo collapse, the mass function, halo bias, and the non-linear matter power spectrum. These effects shape the architecture of the cosmic web and open a rich set of observational probes that will become accessible with upcoming surveys. The results presented here complete the bridge from the linear RDCC framework to the fully non-linear regime and set the stage for future investigations of galaxy formation, cluster physics, and cosmic web topology within the RDCC picture.

References

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